

Can a camera tell which leg needs help?

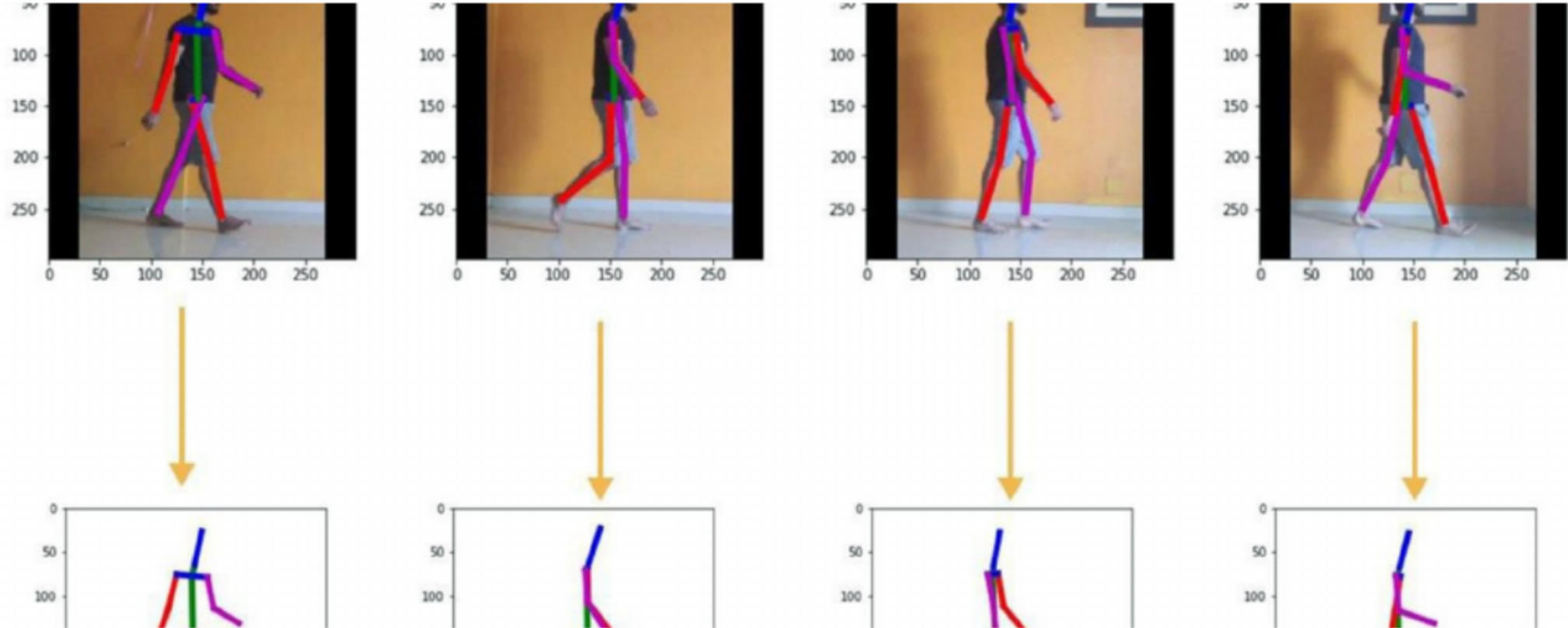
Teaching video models to preserve left-right gait asymmetry

Theodore Mui

Reflection-equivariant S-JEPA

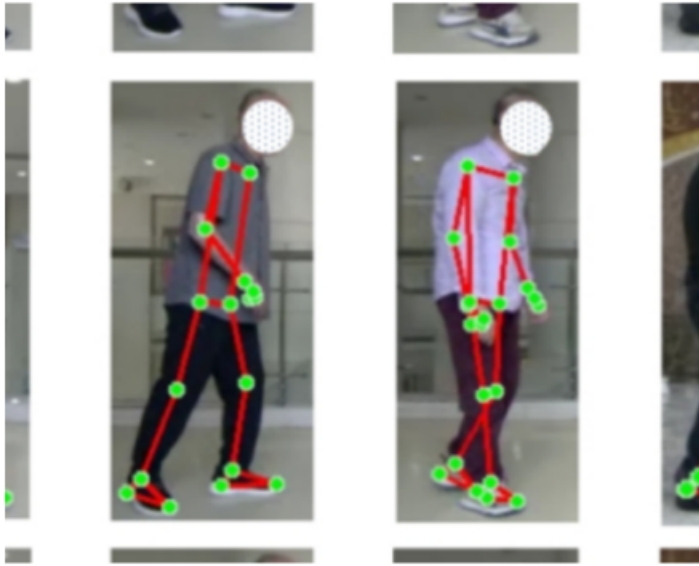


A walk becomes 11 moving points



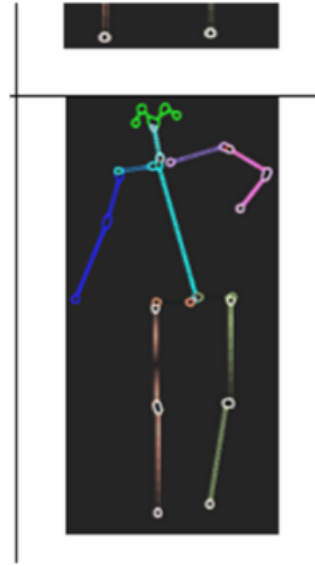
Pose estimation removes appearance and keeps the changing geometry of the body.

Different conditions leave different gait signatures



Stroke

Side-to-side propulsion can become unequal.



Parkinson's

Steps can shorten while posture and arm swing change.



Cerebral palsy

Joint flexion and foot position can follow distinct patterns.

Pose tells us what moved - not what caused it

VISIBLE IN VIDEO

Where each joint went

- Joint positions
- Walking speed
- Timing and posture



NEEDED FOR BIOMECHANICS

How the movement was produced

- Foot-ground contact
- Force and propulsion
- Balance and uncertainty

A faster walk can still hide the wrong kind of recovery

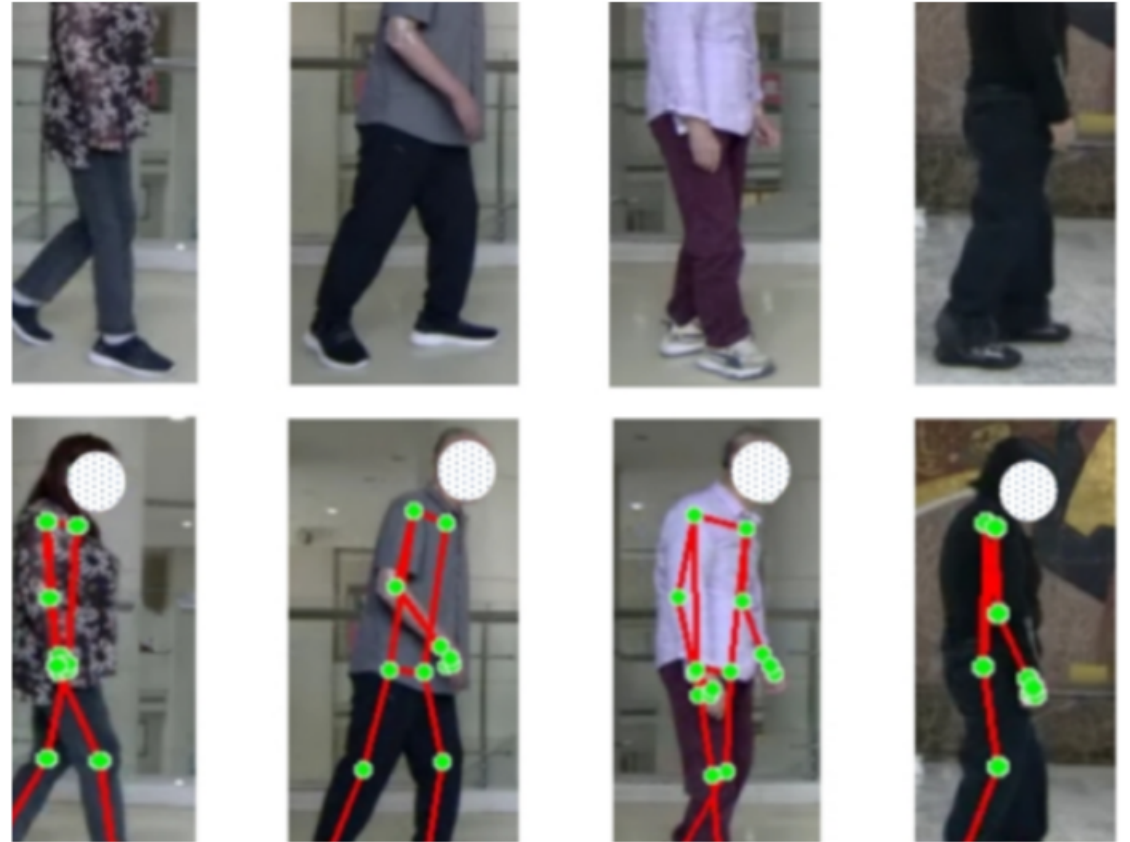
After a stroke, speed may improve because...

the affected leg recovered

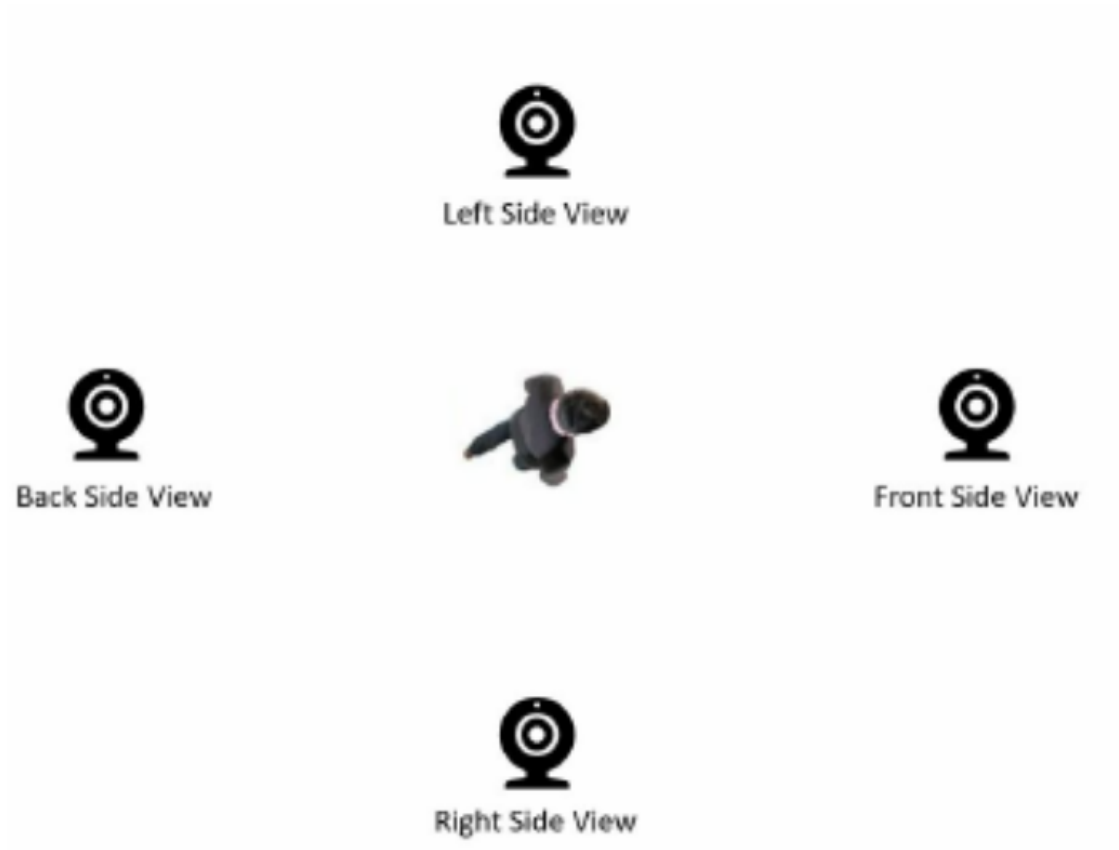
OR

the stronger leg compensated

Those outcomes look similar in speed, but they imply different care.



The opportunity: gait insight from ordinary video



Accessible

Use video captured in clinics or at home.

Consistent

Keep left-right meaning stable across viewpoints and mirrors.

Longitudinal

Separate changes in the person from changes in the camera.

Can a video-derived skeleton reveal which side propels less?

GaitParity tests whether a model built to respect reflection can recover a signed right-versus-left force target for a person it has never seen.

If the input is mirrored, the answer must reverse.

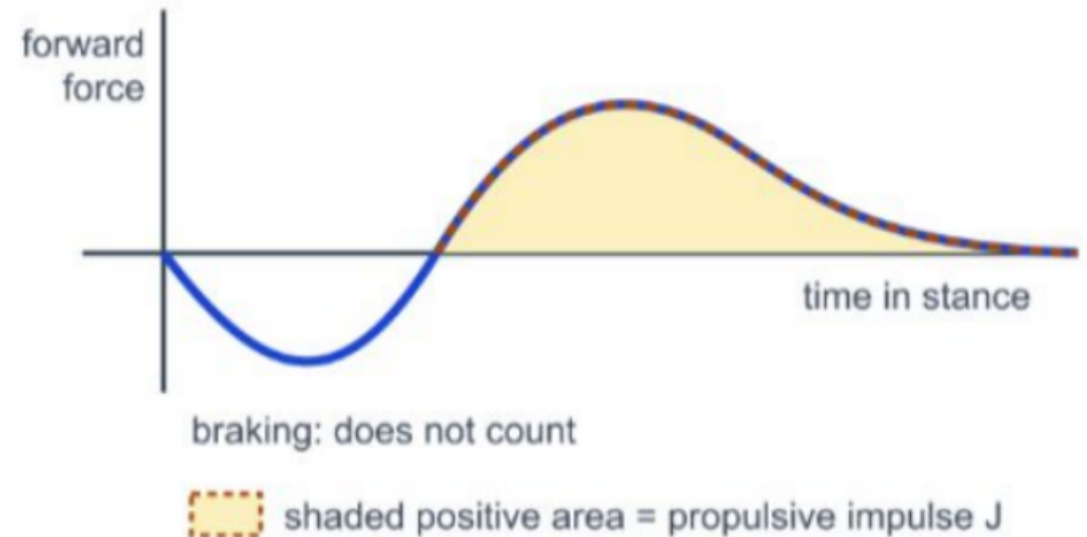
The answer comes from the force under each foot

A force plate measures forward propulsion during stance.

Right impulse versus **Left impulse**

$$\text{signed target} = \ln(JR / JL)$$

The sign identifies the stronger side; the magnitude shows how unequal they are.



A mirror should reverse the answer

A valid anatomical reflection requires three coordinated changes.

1. FLIP THE SIDE AXIS

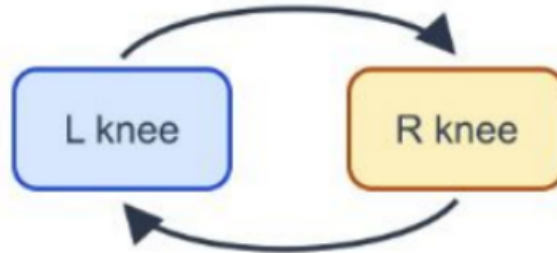
right +0.30 becomes -0.30



Change the coordinate,
not the walking direction.

2. SWAP JOINT NAMES

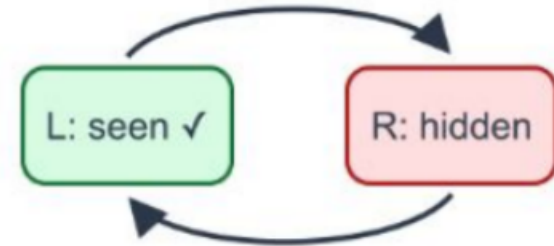
left knee trades with right knee



The geometry and names
must agree.

3. SWAP VISIBILITY

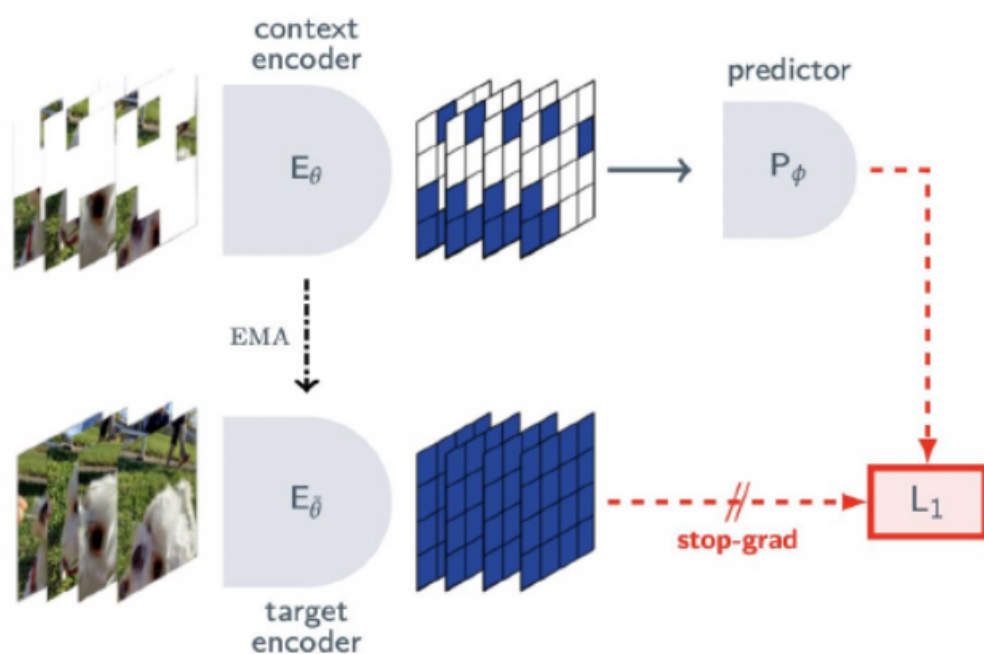
the "seen" flag moves too



Otherwise the model sees
an impossible quality pattern.

The geometry, the joint names, and the visibility flags must all agree.

The model learns by predicting hidden motion



1

Hide pieces

Mask some joints across time.

2

Predict the missing motion

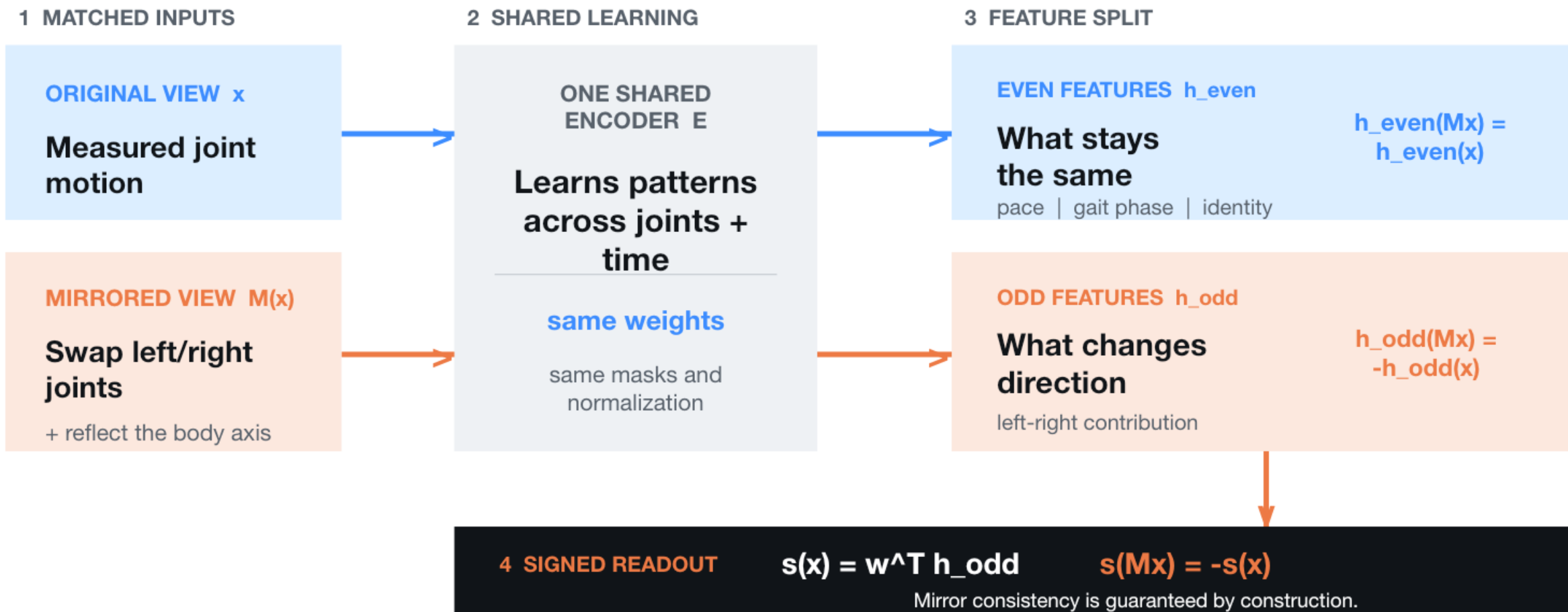
Use the visible context to estimate the hidden representation.

3

Learn structure without labels

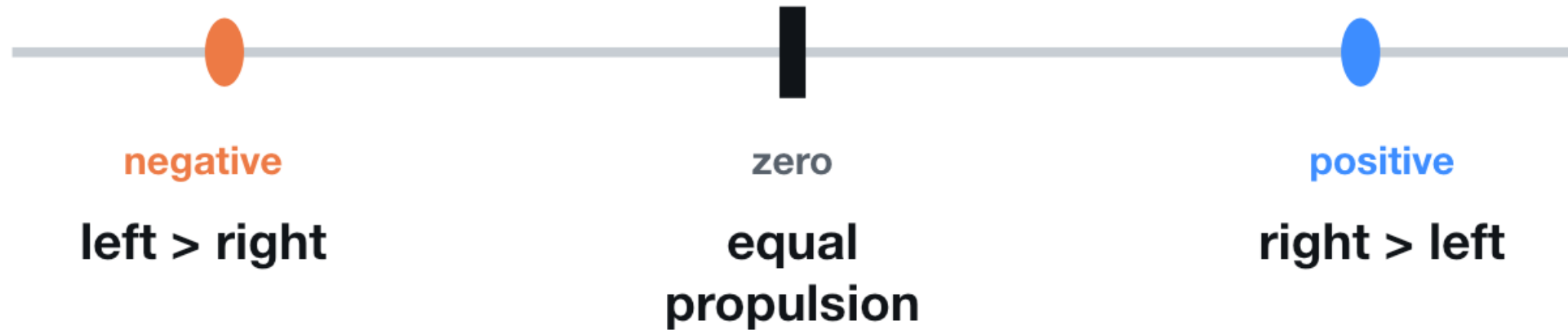
Repeated prediction teaches a compact model of movement.

GaitParity sees every walk in two matched views



One signed score: which side contributes more?

$$\ln(JR / JL)$$



Mirroring the skeleton should multiply this score by -1.

We start with 96 real-world abnormal gait videos

96

gait videos

5

gait categories

Normal
Parkinson's
Stroke
Myopathic
Cerebral palsy



A small, varied set is useful for geometry checks - not yet for clinical claims.

Scale motion learning with 10,941 walking sequences

AMASS supplies diverse 3D motion before force labels are introduced.

10,941

walking sequences

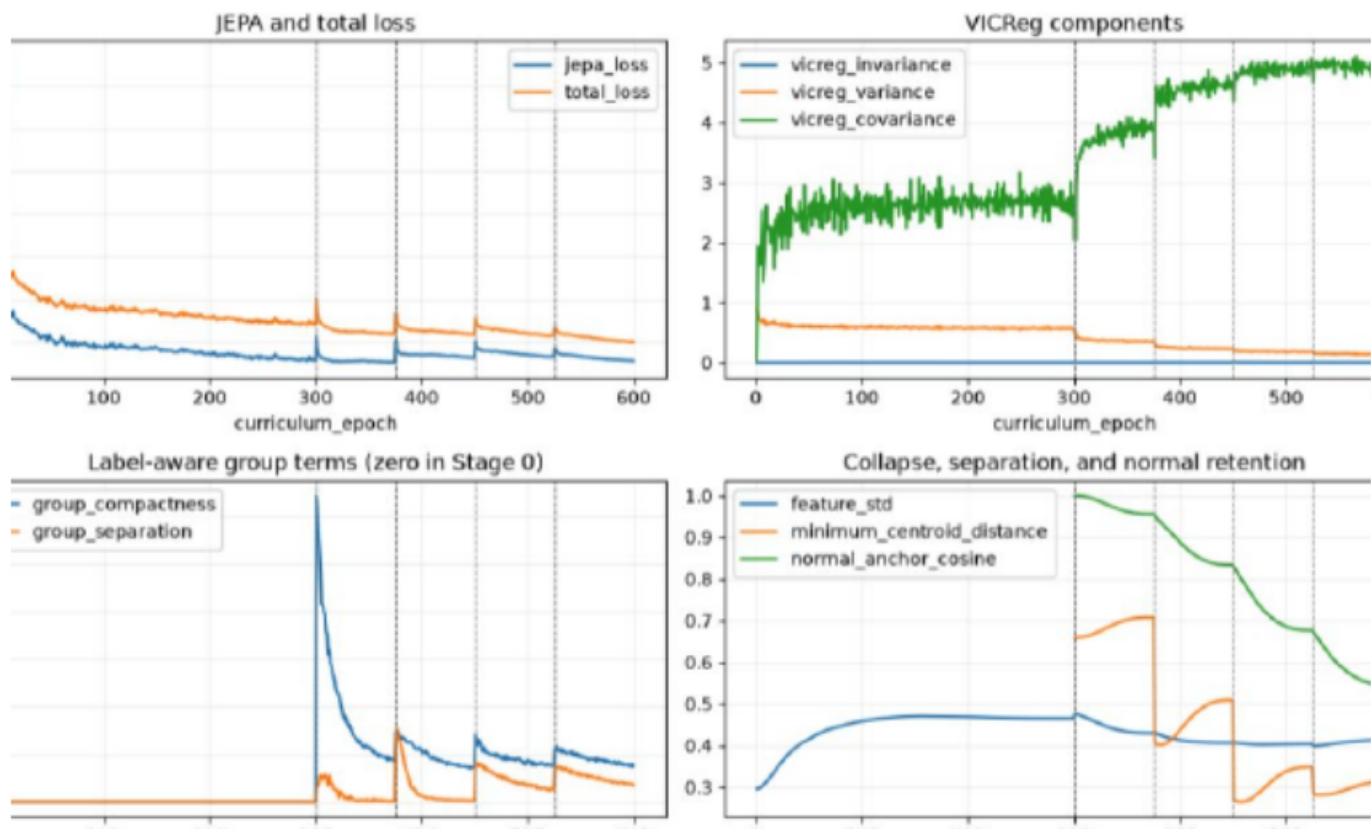
3D

body motion converted to
joint keypoints

**Purpose: learn broad movement structure first,
then specialize to biomechanics.**



Checkpoint 1: the model learned nontrivial motion



Loss fell

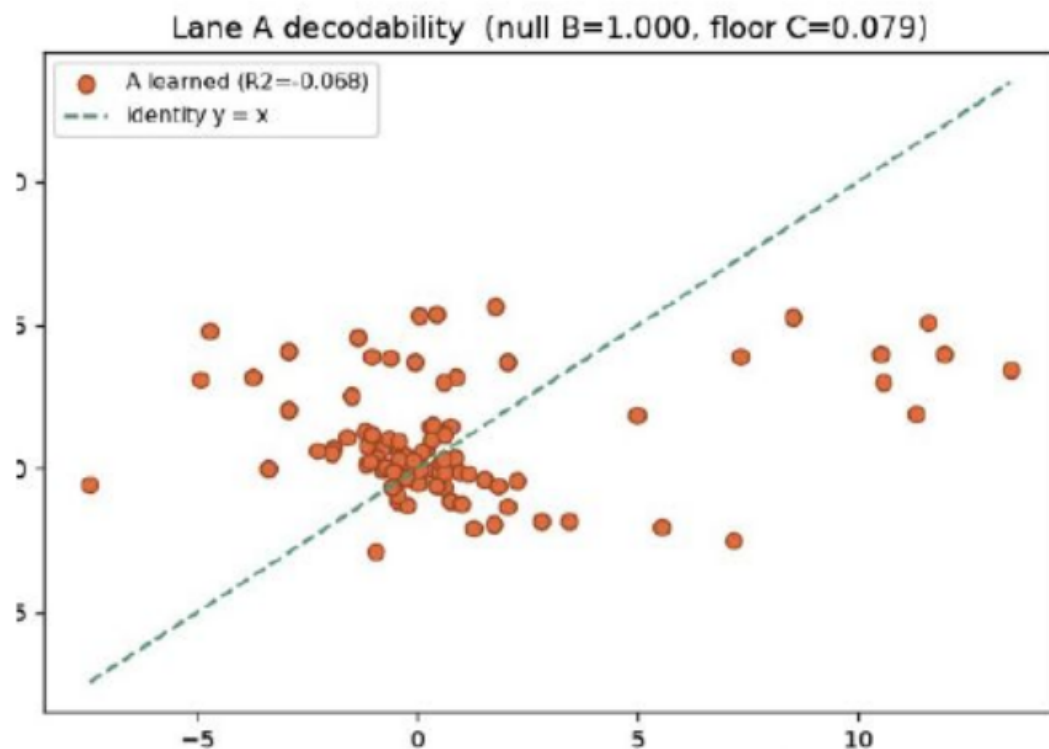
The prediction objective moved in the expected direction.

Features varied

The encoder did not settle on the same representation for every walk.

Meaning: the self-supervised stage learned nontrivial motion structure.

Checkpoint 2: a standard model loses left versus right



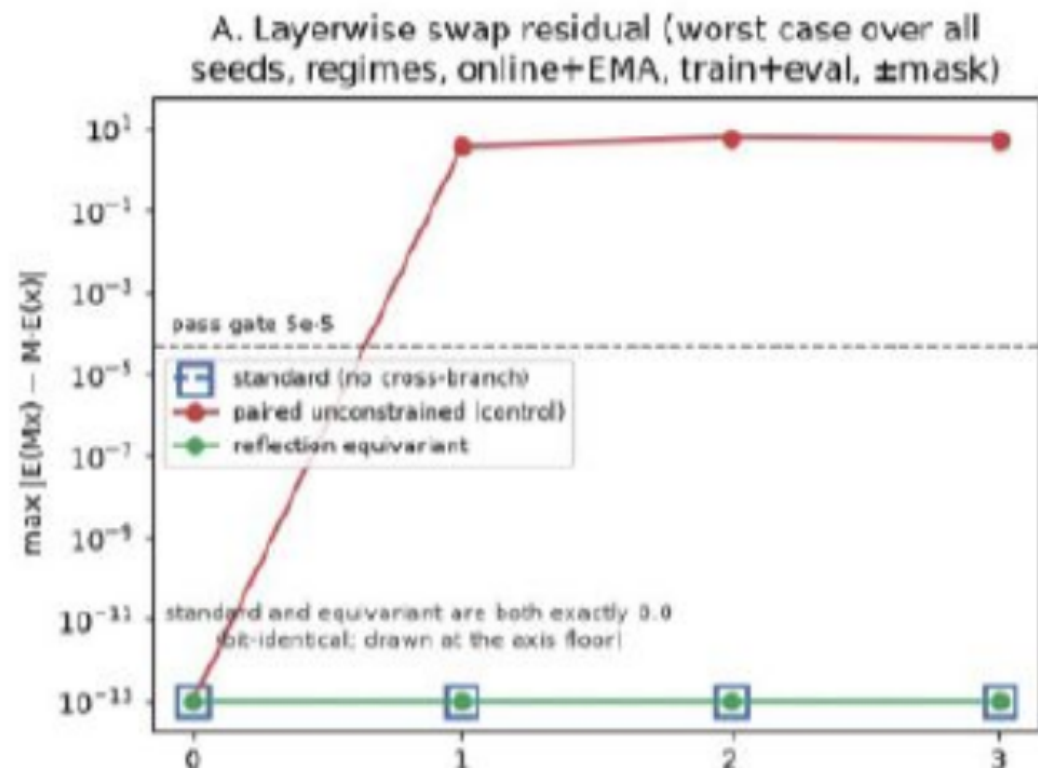
$$R^2 = -0.07$$

No useful signed direction

The baseline can encode motion and still fail to know which side is which.

That failure motivates building symmetry into the representation itself.

The symmetry works. Clinical usefulness is still open.



PASSED

Mirror consistency

The reflection-equivariant encoder follows the intended swap rule exactly in the geometry audit.

STILL TO PROVE

Force prediction on unseen people

Architecture check first; biomechanical validation next.

From tracking movement toward explaining it

1. Pretrain

Finish large-scale AMASS motion pretraining.

2. Ground

Connect motion to force, contact, and uncertainty.

3. Validate

Test signed predictions on unseen people and viewpoints.

Long-term vision: ordinary video that supports biomechanical understanding, not just pose tracking.